

Benchmark Report

# Benchmarking of NQCH's quantum computer

April 30, 2026

## 1. Report of Changes

**Platform:** sinq20  
**Calibration-id:** `ec305fcaf377a05f4fb391a8ee18282bfae18df6`  
**Calibration date:** 2026-04-28 07:58:51  
**Calibration note:** chore(sinq20): update readout parameters  
  
**Experiment-id:** 20260428212203  
**Experiment date:** 2026-04-28 21:22:04  
**Experiment note:** temporary note!!!

**Platform:** sinq20  
**Calibration-id:** `4dc4082f38a53222b3956c22202d32a520d4bc78`  
**Calibration date:** 2026-01-15 02:03:03  
**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9  
  
**Experiment-id:** 20260118161538  
**Experiment date:** 2026-01-18 16:15:38  
**Experiment note:** temporary note!!!

## 2. Version Comparison

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.23  | numpy   | 2.4.4   |
| qibolab      | 0.2.9   | qibocal | 0.2.5   |
| matplotlib   | 3.10.8  | scipy   | 1.17.1  |
| scikit-learn | 1.8.0   | pandas  | 2.3.3   |
| networkx     | 3.6.1   | sympy   | 1.14.0  |
| torch        | 2.11.0  |         |         |

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.22  | numpy   | 2.2.6   |
| qibolab      | 0.2.9   | qibocal | 0.2.4   |
| matplotlib   | 3.10.3  | scipy   | 1.15.3  |
| scikit-learn | 1.6.1   | pandas  | 2.2.3   |
| networkx     | 3.4.2   | sympy   | 1.14.0  |
| torch        | 2.7.0   |         |         |

## 3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



Q7

97.51

Q8

99.83

Q9

99.85

Q10

99.65

Q11

99.69

Q2

99.76

Q3

99.8

Q4

99.09

Q5

99.72

Q6

99.38

Q0

99.47

Q1

99.14

Q17

98.96

Q18

99.72

Q19

99.85

Q12

99.79

Q13

99.79

Q14

98.53

Q15

99.52

Q16

99.26

Q7

98.74

Q8

99.86

Q9

99.76

Q10

98.91

Q11

99.63

Q2

99.82

Q3

99.52

Q4

99.56

Q5

99.46

Q6

99.81

Q0

99.71

Q1

99.76

95.60 96.05 96.50 96.95 97.40 97.85 98.30 98.75 99.20 99.65

1Q Fidelity

88.65 89.85 91.05 92.25 93.45 94.65 95.85 97.05 98.25 99.45

1Q Fidelity

#### 4. Statistics

|                | Average | Median | Min   | Max   |
|----------------|---------|--------|-------|-------|
| T1 (ns)        | 23.1    | 24.5   | 13    | 30.5  |
| T2 (ns)        | 11.4    | 11.7   | 2.18  | 23.9  |
| Fidelity       | 0.951   | 0.993  | 0.855 | 0.999 |
| RO Fidelity    | 0.986   | 0.986  | 0.976 | 0.99  |
| 2q RB Fidelity | -       | -      | -     | -     |
| 2q CZ Fidelity | -       | -      | -     | -     |

|                | Average  | Median   | Min   | Max      |
|----------------|----------|----------|-------|----------|
| T1 (ns)        | 1.28e+04 | 1.23e+04 | 646   | 3.65e+04 |
| T2 (ns)        | 2.36e+25 | 3.68e+03 | 125   | 9.43e+26 |
| Fidelity       | 0.99     | 0.997    | 0.887 | 0.999    |
| RO Fidelity    | 0.794    | 0.777    | 0.777 | 0.927    |
| 2q RB Fidelity | 0.965    | 0.972    | 0.923 | 0.981    |
| 2q CZ Fidelity | 0.975    | 0.974    | 0.967 | 0.991    |

#### 5. Best Qubits Selection

| k-qubits | Best Qubits      | Fidelity |
|----------|------------------|----------|
| 2        | 5, 6             | 0.935    |
| 3        | 5, 6, 10         | 0.929    |
| 4        | 5, 6, 10, 15     | 0.929    |
| 5        | 5, 6, 10, 15, 16 | 0.928    |

| k-qubits | Best Qubits   | Fidelity |
|----------|---------------|----------|
| 2        | 2, 3          | 0.981    |
| 3        | 0, 2, 3       | 0.976    |
| 4        | 0, 1, 2, 3    | 0.970    |
| 5        | 0, 1, 2, 3, 4 | 0.965    |

#### 6. Summary metrics

| Benchmark       | Metric   | Value | Runtime    |
|-----------------|----------|-------|------------|
| Mermin          | Max      | 3.358 | 227.53 sec |
| Grover 2q       | Max prob | 0.874 | 3.13 sec   |
| Grover 3q       | Max prob | 0.225 | 2.44 sec   |
| GHZ state prep. | Success  | 0.903 | 3.17 sec   |

| Benchmark       | Metric   | Value  | Runtime  |
|-----------------|----------|--------|----------|
| Mermin          | Max      | -3.316 | 1.02 sec |
| Grover 2q       | Max prob | 0.89   | 2.36 sec |
| Grover 3q       | Max prob | 0.385  | 2.43 sec |
| GHZ state prep. | Success  | 0.910  | 3.95 sec |

#### 7. Randomized Benchmarking Results

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---------|----------|------------|---------|----------|------------|
| 0       | 0.995    | ± 0.000224 | 10      | 0.996    | ± 0.00012  |
| 1       | 0.991    | ± 0.000662 | 11      | 0.997    | ± 0.000156 |
| 2       | 0.998    | ± 8.9e-05  | 12      | 0.998    | ± 7.26e-05 |
| 3       | 0.998    | ± 7.97e-05 | 13      | 0.998    | ± 9.89e-05 |
| 4       | 0.991    | ± 0.000559 | 14      | 0.998    | ± 5.64e-05 |
| 5       | 0.997    | ± 0.000197 | 15      | 0.998    | ± 0.000118 |
| 6       | 0.994    | ± 0.00018  | 16      | 0.982    | ± 0.00376  |
| 7       | 0.976    | ± 0.00421  | 17      | 0.956    | ± 0.0121   |
| 8       | 0.998    | ± 6.71e-05 | 18      | 0.995    | ± 0.000286 |
| 9       | 0.998    | ± 8.32e-05 | 19      | 0.996    | ± 0.000122 |

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---------|----------|------------|---------|----------|------------|
| 0       | 0.997    | ± 0.000131 | 10      | 0.989    | ± 0.000582 |
| 1       | 0.998    | ± 7.47e-05 | 11      | 0.996    | ± 0.000218 |
| 2       | 0.998    | ± 7.57e-05 | 12      | 0.998    | ± 0.000126 |
| 3       | 0.995    | ± 0.000163 | 13      | 0.998    | ± 7.15e-05 |
| 4       | 0.996    | ± 0.000185 | 14      | 0.985    | ± 0.00183  |
| 5       | 0.995    | ± 0.000147 | 15      | 0.995    | ± 0.000469 |
| 6       | 0.998    | ± 8.34e-05 | 16      | 0.993    | ± 0.000558 |
| 7       | 0.887    | ± 0.0287   | 17      | 0.99     | ± 0.000502 |
| 8       | 0.999    | ± 4.48e-05 | 18      | 0.997    | ± 0.00026  |
| 9       | 0.998    | ± 7.2e-05  | 19      | 0.998    | ± 9.05e-05 |

## 8. Mermin

Mermin's algorithm for 3 qubits.

| Runtime    | Qubits   | Max   |
|------------|----------|-------|
| 227.53 sec | 10, 5, 6 | 3.358 |



| Runtime  | Qubits  | Max    |
|----------|---------|--------|
| 1.02 sec | 0, 2, 3 | -3.316 |



## 9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in [[5, 6]].

| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 3.13 sec | 5, 6   | 0.874    |



| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 2.36 sec | 2, 3   | 0.89     |



## 10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on `siq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in `[[10, 5], [5, 6]]`.

| Runtime  | Qubits   | Max prob |
|----------|----------|----------|
| 2.44 sec | 10, 6, 5 | 0.225    |



| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.43 sec | 0, 2, 3 | 0.385    |



## 11. GHZ state preparation

GHZ circuit with 3 qubits executed on `siq20` backend with 1000 shots.

| Runtime  | Qubits   | Success |
|----------|----------|---------|
| 3.17 sec | 10, 5, 6 | 0.903   |



| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 3.95 sec | 0, 3, 2 | 0.910   |

